Parking Lot Lighting Forecast

Overview
Parking Lot Lighting Forecast is a smart facility energy management project that predicts parking lot lighting electricity usage using sensor-based vehicle count data. The system uses polynomial regression to model the relationship between vehicle occupancy and lighting demand. The results are visualized on a Streamlit dashboard using real-time bar charts, and anomalies in lighting usage are highlighted to support efficient energy management.

Problem Statement

Parking lot lighting systems typically operate at fixed intensity regardless of the number of vehicles present. This results in unnecessary electricity consumption during low occupancy periods. The objective of this project is to forecast lighting demand using vehicle count data obtained from infrared sensors and to detect abnormal lighting usage. This allows lighting systems to adapt based on real-time demand while maintaining safety.

Dataset
The project uses an IoT parking sensor dataset containing the fields created_at, entry_id, field1, field2, and field3. The created_at column represents the timestamp of each sensor reading. The field1 column contains infrared sensor values representing vehicle detection signals such as IR20 or IR40. The vehicle count is extracted from these sensor values. The field2 column represents occupancy status, while field3 contains device information. Lighting electricity usage is estimated from the extracted vehicle count and occupancy flag.

Workflow
Load the dataset and clean column names. Convert the created_at column into datetime format. Extract vehicle count values from the infrared sensor readings in field1. Generate a derived lighting usage variable based on vehicle count and occupancy. Apply polynomial regression to model the relationship between vehicle count and lighting demand. Generate predicted lighting usage values and calculate prediction error. Detect anomalies by comparing predicted and actual lighting usage using a threshold. Display predicted lighting usage through a bar chart and highlight anomalies on the Streamlit dashboard.

Technologies Used

Python, Pandas, Scikit-learn (Polynomial Regression), Plotly, Streamlit.

Installation
Install dependencies using: pip install pandas scikit-learn plotly streamlit

How to Run

Place the parking sensor dataset file in the project directory and run: streamlit run app.py

Key Features

Predicts lighting demand based on vehicle occupancy patterns. Captures non-linear relationships between vehicle count and lighting usage using polynomial regression.Detects abnormal lighting consumption automatically. Provides real-time visualization through interactive bar charts.

Applications
Smart parking infrastructure, adaptive lighting systems, campus energy optimization, safety monitoring, and anomaly detection for maintenance.

Limitations

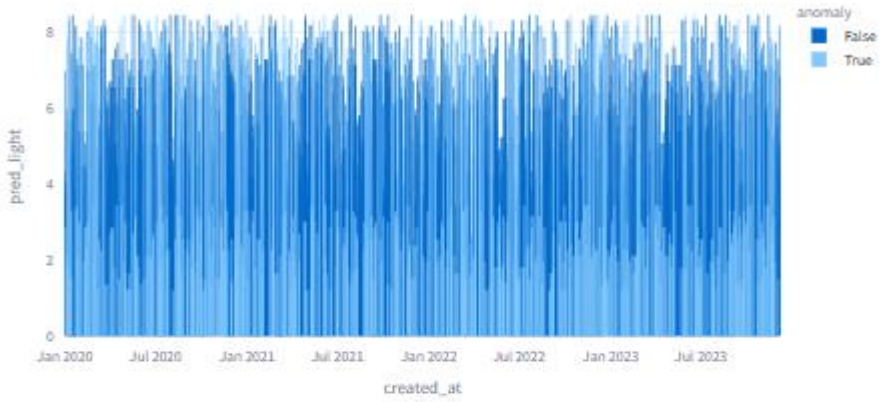
Vehicle count is derived from infrared sensor signals rather than direct counting. Environmental factors such as daylight intensity and weather conditions are not included. Anomaly detection is based on threshold rules instead of advanced machine learning models.

Output

The dashboard displays predicted parking lot lighting usage for each time interval using a bar chart. Anomalies are highlighted when actual lighting usage deviates significantly from predicted values, helping identify abnormal energy consumption patterns.

Parking Lot Lighting Forecast

Parking Lighting Forecast with Anomaly Detection



Anomalies

	created_at	entry_id	field1	field2	field3	vehicle_count	light_kwh	p
3	2020-01-02 17:45:00+00:00	4	IR12	0	2000-0-5T45:162:0	12	1.8	
5	2020-01-03 15:56:00+00:00	6	IR7	0	2000-0-3T45:162:0	7	1.05	
12	2020-01-06 16:30:00+00:00	13	IR46	1	2000-0-5T45:162:0	46	8.9	
14	2020-01-07 08:40:00+00:00	15	IR5	0	2000-0-9T45:162:0	5	0.75	
15	2020-01-08 08:18:00+00:00	16	IR45	1	2000-0-4T45:162:0	45	8.75	
17	2020-01-09 05:46:00+00:00	18	IR4	0	2000-0-4T45:162:0	4	0.6	
19	2020-01-10 07:16:00+00:00	20	IR50	1	2000-0-9T45:162:0	50	9.5	
22	2020-01-12 17:42:00+00:00	23	IR44	1	2000-0-5T45:162:0	44	8.6	
24	2020-01-13 17:31:00+00:00	25	IR3	0	2000-0-3T45:162:0	3	0.45	
28	2020-01-15 19:51:00+00:00	29	IR15	0	2000-0-5T45:162:0	15	2.25	

Future Enhancements

Integrate ambient light sensors and weather data. Use advanced anomaly detection models. Enable real-time IoT data streaming. Implement automatic lighting control based on predicted demand.

Conclusion

The project demonstrates that polynomial regression can effectively model the relationship between vehicle occupancy and parking lot lighting demand. By combining forecasting, anomaly detection, and dashboard visualization, the system enables adaptive lighting management and improved energy efficiency in parking facilities.